Machine Translated by Google

SERIES OF TECHNICAL NOTES ON THE IMPACT OF NATURAL GAS DEFICIT AND THE INCREASE IN PRICES FOR END USERS: GENERAL PRESENTATION

Note 4. Short and medium term public policy and regulation for natural gas

Authors:

Juan Benavides

Sergio Cabrales

November 4, 2025

FEDESARROLLO
Centro de Investigación Económica y Social

Machine Translated by Google

Technical notes series on the impact of the natural gas shortage and price increases for end users: general overview

Due to a combination of past and present public policies that have discouraged investment in the hydrocarbon sector, increased transaction costs for the exploration and exploitation of conventional and unconventional domestic natural gas, inflexibilities in long-term gas contracts, lack of construction of strategic storage facilities and distance charges in the transportation system, Colombia faces a structural deficit and possible shortage of natural gas in the short term.

Fedesarrollo presents four technical notes on the impacts of the natural gas deficit due to the reduction in domestic supply, the difficulties in expanding import capacities in the short term, and the increase in prices for end users due to import needs.
These notes aim to provide elements for decision-making within the State and to inform the public with public data and arguments guided by the general interest.

The titles of the technical notes are as follows:

Note 1. Increase in the price of residential natural gas in scenarios of greater imports.

Note 2. Fiscal cost of increased subsidies due to rising natural gas prices.

Note 3. Macroeconomic costs of reducing the supply of natural gas.

Note 4. Short and medium term public policy and regulation for natural gas.

Note 5. Energy poverty due to reduced supply and increased price of natural gas.

Machine Translated by Google

Note 4. Short and medium term public policy and regulation for natural gas

Juan Benavides and Sergio Cabrales1 .

Objective and summary of recommendations

This document summarizes the results of the previous Notes in the series, proposes energy policy principles and discusses the role of gas in Colombia, diagnoses and categorizes the problems that need to be solved in the natural gas chain, and formulates recommendations.

The 2026-2030 presidential administration faces the following tasks that require great agility, persistence, and political will:

ÿ Promote the orderly entry of liquefied natural gas (LNG) import facilities of at least 400 Giga BTU per day (GBTUD).
ÿ Accelerate the incorporation of proven reserves, increase production and enable fracking.
ÿ Reduce government *take* from the hydrocarbon sector.
ÿ Rationalize and refocus public support (subsidies) to the sector.
ÿ Accelerate and limit the environmental licensing process and prior consultations.
ÿ Migrate to a national stamp scheme to remunerate the transportation of natural gas by networks.
ÿ To make the contractual conditions for the import and marketing of gas more flexible.
ÿ Promote the development of projects for the strategic storage of natural gas and regulate their remuneration.

General results of the Technical Notes on natural gas2

ÿ Proven natural gas reserves in Colombia have fallen by 64% in 11 years, from 5727 GPC in 2012 to 2373 GPC in 2023. Except for 2021, produced reserves have not been replaced.
ÿ The 2022 tax reform increased the State's share of the profits generated by hydrocarbon projects, raising the *government take3* from 63% to more than 80% (Corficolombiana 2022).

ÿ On December 1, 2024, after 45 years, Colombia lost its self-sufficiency to cover the essential demand for natural gas, residential and commercial sectors, vehicular natural gas (CNG), refineries and compressor stations.

ÿ Projections of the supply and demand balance indicate that, towards the end of 2026, the supply of natural gas, both domestic and imported, would not be sufficient to cover the

---

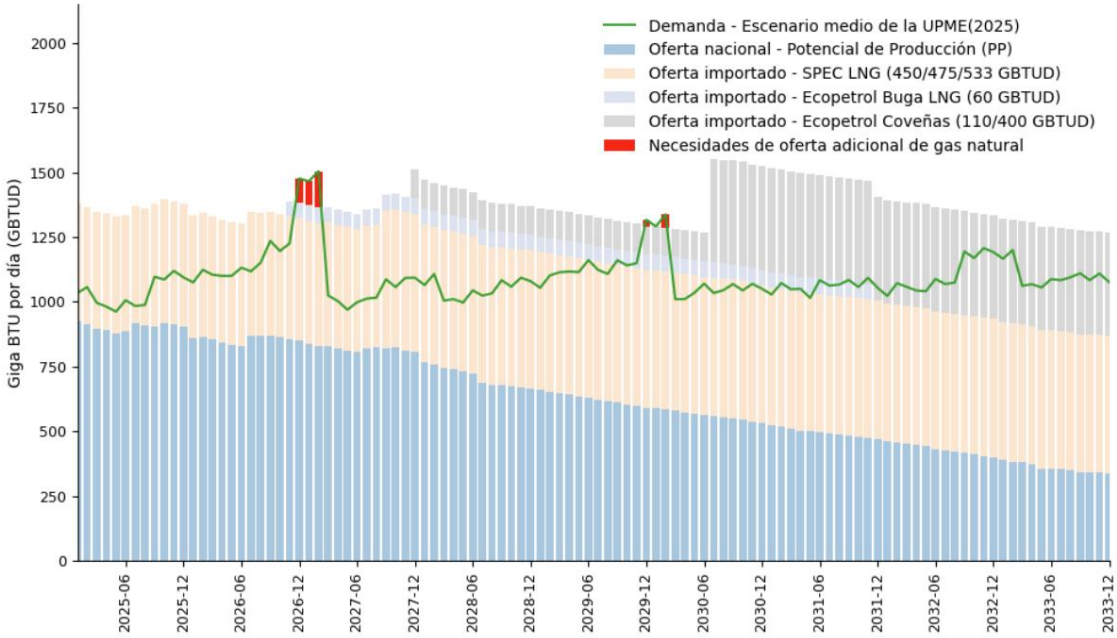
1 Disclaimer: The opinions expressed in this publication are those of the authors and should not be attributed to Fedesarrollo.

2 Cabrales and Benavides (2025A and 2025B); Ramírez et al (2025).
3 The *government take* is the part of the real economic rent generated by an investment during the life of a project that a government collects, calculated in present value of the cash flows it will receive.

Machine Translated by Google

projected demand by UPME (deficit in red bars in Graph 1), despite including the new supply of imported gas (Buga and Coveñas).

Figure 1. National balance between national supply (Production Potential, PP) and imports versus projected demand



Source: Prepared by Fedesarrollo with data from the Ministry of Mines and Energy, UPME, SUI-SSPD, SEGAS and Concentra.

ÿ If half of the supply came from imported gas, the tariff increase could reach 44.6% in Bogotá, 44.9% in Medellín, 45.8% in Bucaramanga, 30.3% in Cali and 11.7% in Barranquilla, respectively.

ÿ In a scenario where the basket is composed exclusively of imported gas, the increases could be 89.3% in Bogotá, 89.7% in Medellín, 91.5% in Bucaramanga, 60.7% in Cali and 23.4% in Barranquilla.

ÿ Imported gas, besides being insufficient to meet the local deficit due to the limited capacity of the regasification plant in La Cartagena, has prices that can be up to three times higher than those of national gas because it does not have long-term contracts.

- Regulated gas users grew by 218.7% between 2005 and 2024, going from 3.6 to 11.5 Millions. By 2025, 78% of households in municipal capitals will use gas. 59.8% of users belong to socioeconomic levels 1 and 2, which receive subsidies of 50-60%, while levels 5 and 6 (5.3%) contribute 20%.

ÿ In 2005, subsidies to households in strata 1 and 2 represented COP $0.19 trillion, compared to COP $1.41 trillion today, which is equivalent to an increase of 642% in the last 20 years.

ÿ If half of the supply came from imported gas, the tax burden would increase by COP $0.36 billions annually.

ÿ Residential rates could increase between 11.7% and 44.9% in five major cities with imports at 50% of demand and

Machine Translated by Google

Without the addition of domestic reserves, between 2025 and 2031, GDP would grow 0.1 percentage points less due to rising gas prices, resulting in a cumulative loss of $11.2 trillion. Industrial growth would fall by 0.6 percentage points, trade by 0.1 percentage points, and GDP by 0.1 percentage points annually.

ÿ From 2032 onwards, without new reserves, tariff increases and restrictions would deepen losses, with a reduction of 1.4 points in industrial growth and 0.3 in trade, and GDP would fall by up to 0.4 points in the medium term, from 2.9% to 2.5%.

ÿ On the external front, the current account deficit would increase by 0.2 percentage points of GDP per year from 2026 onwards, due to increased gas imports.

Colombia is losing its appeal for hydrocarbon investment. Major oil companies like ExxonMobil and Shell have left and no longer have any significant projects planned in the country.

Principles of energy policy and natural gas development

The country must follow five principles of energy policy (Benavides 2025):

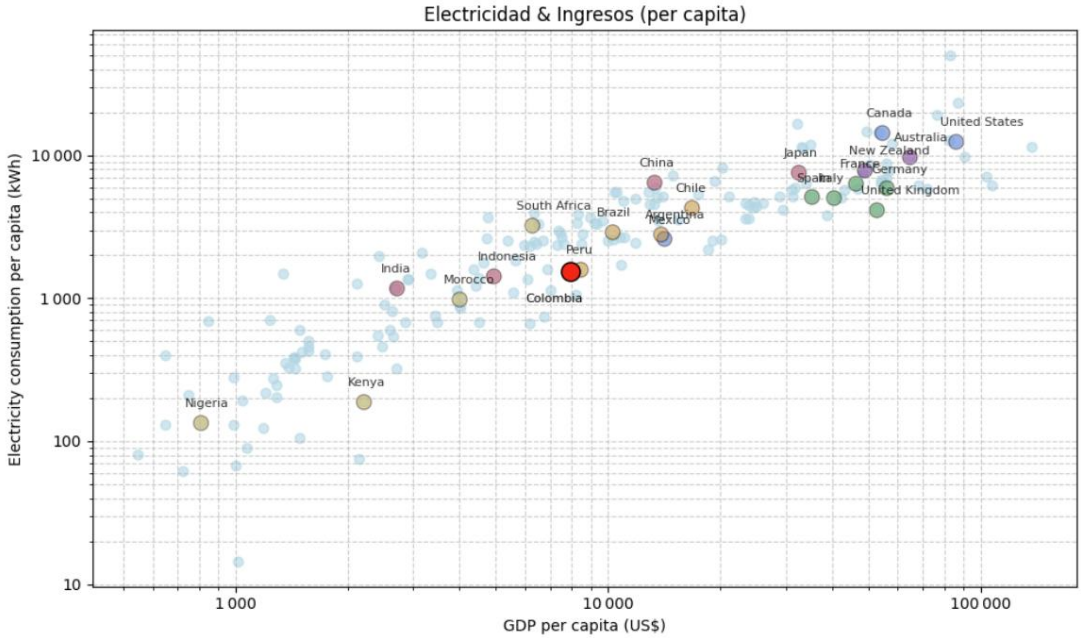
Correcting the Mistakes of the 'Energy Transition'. Fortunately, the notion of 'energy transition' is undergoing a process of revision due to the difficulty of quickly and cost-effectively eliminating dependence on hydrocarbons in the major commodities upon which the modern economy relies (cement, steel, plastics, and fertilizers), and in processes that demand high energy density per unit volume (interoceanic shipping, furnaces, and boilers). Colombia lacks the financial resources and the emissions weight to embark on massive decarbonization investments. *In 30 years, not all final energy consumption will be electrified, nor will all electricity be produced from renewable sources.*

Do not place the burden of greenhouse gas reductions on the energy sector. Colombia's energy production and consumption should not bear the brunt of the greenhouse gas (GHG) emissions reductions to which it committed at COP26, because the majority of these emissions come from agriculture, deforestation, and land-use change (approximately 35% energy versus 55% AFOLU (Figure 2)). In terms of per capita emissions, Colombia is the target that developed countries aspire to reach within 30 years.

Subordinate energy policy to economic growth. As Figure 2 shows, there are no wealthy countries that do not have high per capita electricity consumption. And as Figure 3 shows, energy consumption is correlated with the Human Development Index. Colombia must escape the middle-income trap (WB 2024) at a faster rate than its population is aging. Energy consumption must be increased from the meager 43 gigajoules (GJ) per capita per year in 2024, far below the global average of 73 GJ/year, by more than doubling to sustain GDP growth of approximately 4% annually. Hydrocarbon production must be increased to support tax revenues.

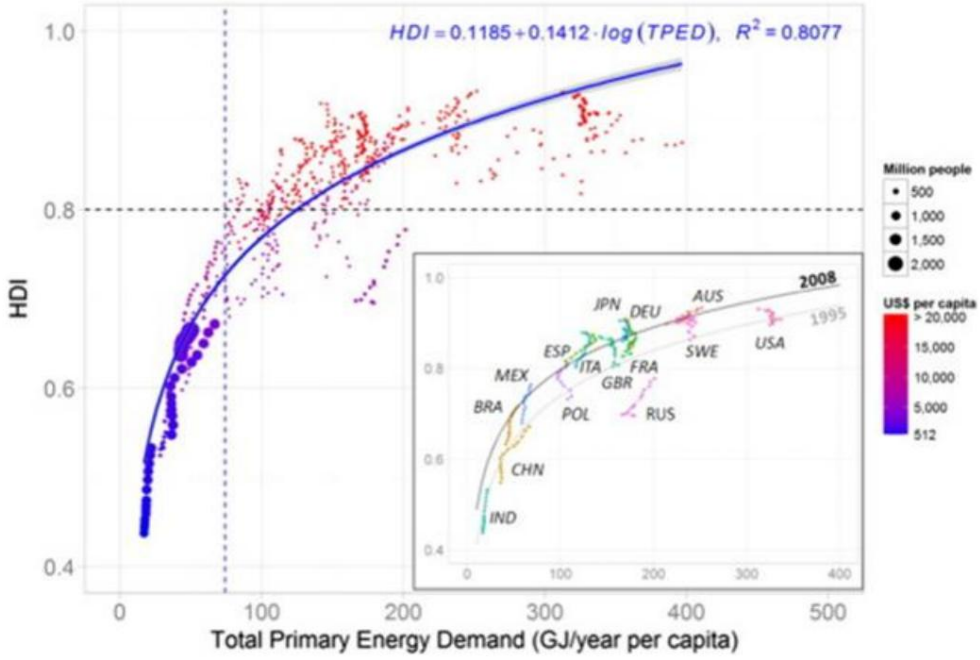
Machine Translated by Google

Figure 2. Correlation between electricity consumption and income per capita



Source: Prepared by Fedesarrollo with data from the World Bank4 and the International Energy Agency (IEA) 5

Figure 3. Per capita energy consumption and Human Development Index



Source: Arto et al (2016).

Expand the energy supply as a portfolio that values the attributes of each source. The welcome reductions in unit costs of non-conventional renewable energy sources do not cover the needs for continuity, density in industry, and, when invested without

[4] https://data.worldbank.org/indicator/NY.GDP.PCAP.CD
[5] https://www.iea.org/data-and-statistics/data-tools/energy-statistics-data-browser

Machine Translated by Google

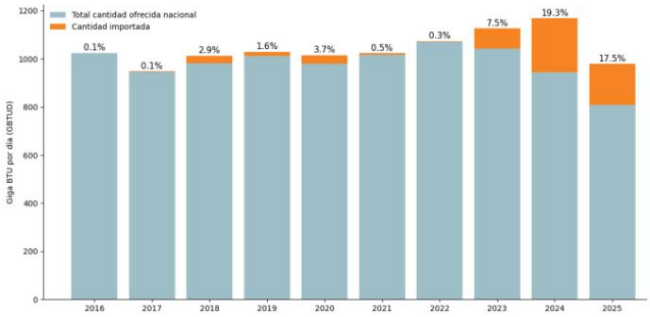
Battery backup and grid integration can create various problems in electricity networks. New energy supply planning methodologies available internationally balance costs, service continuity, diversification, and environmental contributions. In line with principle 1, relying on a single energy source is unrealistic, and investments must be profitable without subsidies.

Betting on natural gas as a permanent energy source. The idea of assigning natural gas a shameful and temporary role in the 'energy transition' (understood as decarbonization) has gained traction, even among industry players. This is a mistake that can have repercussions through the following channels: (i) through reduced tax revenues; (ii) through the additional costs of excluding natural gas from electricity generation to preserve the same generation levels and greater investment needs in grid security; and (iii) through the current higher total cost of ownership of cleaner technologies in the industry.

Problems in the natural gas sector6

Increased risk of shortages. In the next 1-3 years, Colombia will face shortages of electricity and natural gas. Such a simultaneous risk has not occurred since 1994. Rationing electricity and gas would cost 1.5% of GDP growth, result in the loss of at least 260,000 jobs annually, and push an additional 250,000 people into poverty (Fedesarrollo 2025). From day one, the current government declared a moratorium on hydrocarbon exploration in new blocks. By the end of 2024, Colombia will have lost its self-sufficiency in meeting the "essential demand" of natural gas in the residential, commercial, and vehicular (CNG) sectors, as well as in refineries and compressor stations, with its domestic production. SPEC LNG no longer only imports gas for the electricity sector on an occasional basis. So far in 2025, 17.5% of the natural gas consumed has been imported (Graph 4) and in 2024, 23.8% of the total gas offered was imported, compared to 8.1% in 2023.

Figure 4. Average annual gas supply in Colombia (2016-2025)



Source: Prepared by Fedesarrollo with data from the Ministry of Mines and Energy, UPME, SUI-SSPD, SEGAS and Concentra.

6 Benavides 2025; Cabrales and Benavides 2025A and 2025B.

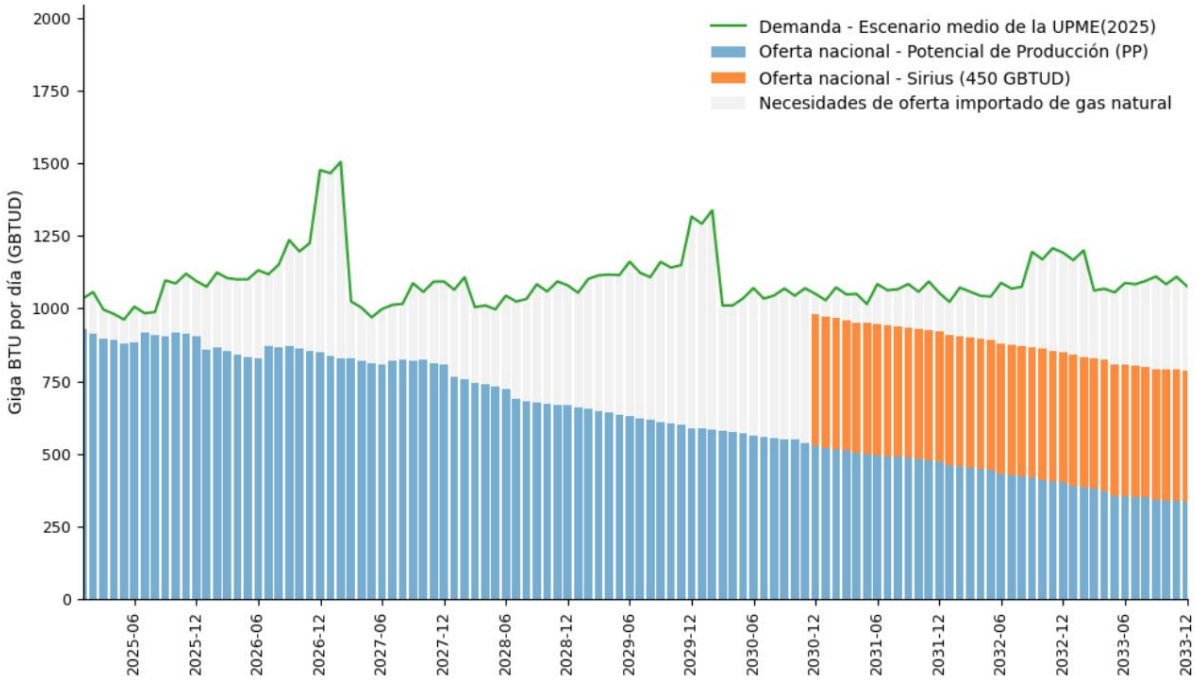
Machine Translated by Google

Projections of the balance between supply and demand, based on official information, indicate that, towards the end of 2026, the supply of national and imported natural gas would not be sufficient to cover the demand projected by the UPME in the event of an El Niño Phenomenon, and that increasing additions to supply are required to avoid rationing in the summer of 2030, and permanently from 2031 onwards.

Sirius-2 is an offshore gas project near Santa Marta, jointly owned by Ecopetrol and Petrobras. It has a potential supply of 450 million cubic feet per day for 10 years, approximately 45% of current national demand. According to Ecopetrol, it is the largest gas discovery in the country's history (Ecopetrol 2024). Its entry into operation requires US$4 billion to reach production. To avoid a one-year delay in the schedule, the project must resolve 122 prior consultations and obtain 3 environmental licenses, a process that began in April 2025. Based on the logic and incentives of the National Environmental Licensing Authority (ANLA) and the Ministry of Environment and Sustainable Development (MADS), this critical project would not come online until 2030, a year after which, without the addition of domestic reserves or imports, a structural supply deficit would solidify.

(Graph 5).

Figure 5. National balance between national supply (Production Potential, PP) and Sirius-2 versus projected demand



Source: Prepared by Fedesarrollo based on data from the Ministry of Mines and Energy, UPME, SUI-SSPD, SEGAS, Concentra and Petrobras.

The subsidy scheme is unsustainable. Between 2005 and 2024, subsidies for socioeconomic levels 1 and 2 increased by 642%, while contributions from levels 5 and 6 only increased by 157%, generating a fiscal deficit that rose from COP $0.17 trillion to COP $1.25 trillion. NATURGAS estimates that the National General Budget (PGN) will require between COP $1.3 and COP $1.4 trillion in natural gas subsidies in 2025-26. With 50% of gas imported, the fiscal burden could increase by an additional COP $0.36 trillion, or up to COP $0.72 trillion if it is 100% imported.

Machine Translated by Google

imported. After adding the contributions of regulated and unregulated users, the overall balance between subsidies and contributions for the entire system is shown in Chart 6.

Chart 6. Overall balance between contributions and subsidies (2025 pesos)



Source: Prepared by Fedesarrollo based on data from the Single Information System (SUI).

Licensing and community consultations can delay the implementation of critical projects for years. This process is not aligned with national infrastructure and energy investment needs. There is no proportionality between the administrative delays and the protection of thousands of residents near the project site, versus the urgent need to deliver benefits to millions of people. The procedural requirements are similar for both complex and simple projects. Community consultations are based on international legislation that grants them veto power over project development, based on the presumption that every project entails significant harm to communities and the environment.

Natural gas regulation does not promote competition in production or the commercial integration of the two submarkets. The gas transportation network remains radial after 30 years of regulatory practice. Distance-based transportation charges have entrenched two production submarkets (Coastal and Inland) with a dominant player in each, restricting the entry of new domestic production and competition. Term transportation contracts and regulated charge pairs hinder investment in pipelines fed by small fields. A spot market has not been established. There has been no progress toward greater use of financial instruments (physical transactions prevail). Strategic storage facilities have not been introduced.

Machine Translated by Google

Policy Recommendations7

Promote the orderly commissioning of natural gas import facilities with a capacity of at least 400 gigabits per day. The sum of domestic production and imports is insufficient to meet the country's essential demand or for gas-fired thermal power generation. Even if the Sirius project were to begin development today, by 2030 it would not be enough to cover the structural deficit, and we would continue to import liquefied natural gas. Numerous natural gas import initiatives are at various stages of development. Under these circumstances, holding a two-attribute auction (cost and commissioning time) for imports is the best way to minimize the risk of rationing. Investment in imports should not be state-owned by definition. Competition among private entities will ensure the best combination of price and delivery speed. An additional benefit is that lower import CAPEX investment will cap production prices, benefiting households, industry, and industrial expansion, as well as enabling the necessary expansion of weather-independent power generation.

Accelerate the incorporation of proven reserves, increase production, and enable *fracking.* The Ministry of Mines and Energy (MME) will lead the acceleration of production from Sirius, La Belleza, Floreña, and Arrecife, as well as the expansion of Gibraltar. The National Hydrocarbons Agency (ANH) must promote *fracking* 4.0, a technology with a mature learning curve worldwide and whose advancements reduce its environmental impacts and risks. Fracking is a short- to medium-term solution. Colombia has draft fracking E&P contracts available at the ANH and some contracts already signed (currently suspended). It has firm technical and environmental regulations and a positive opinion from the Council of State on them. Investment decisions in fracking should be based on cost-benefit and risk analyses on a project-by-project basis, not on pilot projects. By combining offshore projects, such as Sirius-2, with fracking, natural gas self-sufficiency could be achieved by the beginning of the next decade.

Reducing the *government take* on natural gas. The ANH, after a working group with the MME and the MHCP, will reduce the *government take* on hydrocarbons (which would reach 80%) to less than 60% to adjust it to the risk conditions and size of Colombia's gas fields, in order to compete in attracting investment with Mexico, Argentina, Brazil and Guyana, at least.

Rationalize public support (subsidies) for the sector. It's not enough to simply pay what is owed; the focus should be on criteria better than socioeconomic status. A transitional solution is to use consumption-based criteria, due to its ease of application, while a definitive solution is developed that uses more sophisticated criteria, primarily to reduce inclusion errors. The number of subsidy recipients, currently 60% of households, must be reduced.

Accelerate and streamline the environmental licensing and prior consultation process. The model for Projects of National and Strategic Interest should be reinstated, which can be achieved by expanding the scope of the Intersectoral Commission for Infrastructure Projects (CIIPE) to include energy projects. The CCIPE will focus on facilitating the inclusion of the most critical supply projects. To this end, it will (i) simplify the environmental licensing process; and (ii) replace prior consultatio

---

7 Benavides (2025).

Machine Translated by Google

with individual communities through an aggregated process of peremptory deadline; (iii) will focus compensations, if required, on local public goods (community projects for health, education, drinking water and digital connectivity).

Migrate to a national stamp tax scheme to remunerate the transportation of natural gas through pipelines. To ensure that new gas injections reach their destination and maintain the profitability of assets, CREG will (i) adopt outstanding network charges (some more than 5 years overdue); (ii) will resolve requests for the aggregation of sections; (iii) will introduce the collection of transport by stamp in networks to encourage the entry of domestic reserves that are currently not competitive with the charge by distance; (iv) will establish peremptory deadlines for Promigas to put into operation the bidirectionality of Barranquilla-Ballena and Cartagena-Jobo, and the expansion of the Ballena-La Mami section, and TGI to put into operation the bidirectionality of Vasconia-La Belleza; and (v) will ensure that the interconnection between the systems of Promigas and TGI in the Ballena node is at the maximum exchange capacity in that node (300 MMSCFD).

Making Contractual Conditions for Gas Imports and Marketing More Flexible. With the aim of strengthening the country's energy security and promoting a more competitive environment in the natural gas market, it is necessary to make the contractual conditions associated with import and marketing processes more flexible. By allowing greater flexibility in aspects such as deadlines, volumes, and prices, the possibility of accessing imported natural gas under more favorable conditions is opened, guaranteeing an efficient, competitive, and sustainable supply for the various sectors of national consumption.

Promote investment in storage. In practice, regasification terminals can shape the first phase of detailed development for strategic onshore storage projects. In a second phase, depleted oil and gas fields should be evaluated, along with their compensation mechanisms. This type of storage infrastructure is widely used for seasonal system supply, to meet peak demands, or to address contingencies.
Restrictions or scheduled maintenance, firstly. Additionally, storage can support liquidity formation in a secondary gas market.

References

Benavides, J. (2025). Strategy 2026-2030 for electricity and natural gas in Colombia. Fedesarrollo.

Benavides, J. and S. Cabrales. (2022). Energy transition in Colombia: Policy, costs of accelerated carbon neutrality and the role of natural gas. Fedesarrollo.

Cabrales, S. and J. Benavides. (2025A). Note 1. Increase in the price of residential natural gas in scenarios of greater imports. Fedesarrollo.

Cabrales, S. and J. Benavides. (2025B). Note 2. Fiscal cost of increased subsidies due to rising natural gas prices. Fedesarrollo.

Corficolombiana. (2022). Special Report. Tax Reform: Analysis and Expected Impact.

Machine Translated by Google

https://investigaciones.corfi.com/macroeconomia-y-mercados/informes-especiales/reforma-tributaria-analisis-e-impacto-potencial/informe_1225728

Fedesarrollo. (2025). How are the energy balances in Colombia? Perspectives and challenges. Presentation by the director of Fedesarrollo at Acolgen, May 2025.

WB. (2024). The middle-income trap. World Development Report.

https://www.worldbank.org/en/publication/wdr2024